

A General Definition of *Ramification Bounded by r*

Proposed edits to §2.1–§2.2 — review copy

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Abstract

This is a standalone review document. It is not part of the thesis and is not included by `main.tex`. It records four proposed edits whose purpose is to make the phrase “ramification bounded by r at (Z, ξ) ” — used in the statements and proofs of the stability lemmas of [Section 2.2](#), and applied at the generic point of an exceptional divisor — into a defined notion. Material that is [new or changed](#) is set in blue. Theorem and definition numbers inside this document are local to it and bear no relation to the numbering in `main.pdf`.

The gap

[Section 2.2](#) opens in the right generality: a locally Noetherian scheme X , a dense open $U \subset X$, and a prime divisor (Z, ξ) of $X \setminus U$ whose local ring $\mathcal{O}_{X, \xi}$ is a discrete valuation ring. In that generality it defines only *unramified* and *tamely ramified* ([Definition 38](#)). It then says “now back to our original assumptions” and descends to a closed point P of the curve C , where it defines *ramification bounded by r* and the character dictionary $\rho_P : G_{\widehat{K}_P} \rightarrow G$.

Two things are therefore used but never introduced in the generality in which they are used:

1. the phrase $\mathcal{P}$ *has ramification bounded by r at (Z, ξ)* , which is the hypothesis and the conclusion of the contracted-product lemma and of the three stability lemmas that follow it;
2. the character reformulation $\rho_\xi(G_{\widehat{K}_\xi}^r) = 1$ at a general ξ , which is what the proofs of those lemmas actually manipulate — each of them begins “let $A = \widehat{\mathcal{O}_{X, \xi}}$, $K = \text{Frac}(A)$, and let $\rho_i : G_K \rightarrow G$ be the associated characters”.

This is not a cosmetic gap. The lemmas are invoked at $\eta_{\mathfrak{n}}$, the generic point of the exceptional divisor $E_{\mathfrak{n}} \subset X_{\mathfrak{n}}$ — exactly the case the closed-point definition does not cover.

The design question, and the choice made here

The upper-numbering filtration is currently constructed in [Section 2.1](#) under the standing hypothesis *complete, equal characteristic p , perfect residue field*. At $\eta_{\mathfrak{n}}$ the residue field is $k(E_{\mathfrak{n}}) = k(\mathbb{P}^{d-1})$, a function field, hence *imperfect* in characteristic p . So the general definition cannot simply cite that setup as it stands. Three ways out:

1. Require $\kappa(\xi)$ perfect. Cleanest to state — and excludes the only case we need.
2. Develop ramification theory for imperfect residue fields (Abbes–Saito). Correct, and out of all proportion to the need.
3. [Weaken the standing hypothesis from “perfect residue field” to “separable residue extension”.](#)

Option 3 is what is proposed below, for three reasons. It is Serre’s own standing hypothesis in [\[1, Chapter IV\]](#); every construction in [Section 2.1](#) goes through under it verbatim, since the only place perfectness is used is to produce a monogenic generator of B over A ; and it is already what

the proofs verify by hand — the unramifiedness argument in the Artin–Schreier case establishes precisely that each field factor has “ramification index 1 and separable residue extension”.

The cost is that residual separability becomes a hypothesis one must carry and check. That is honest: it *is* a hypothesis at an imperfect residue field, and pretending otherwise is what produced the gap in the first place.

Edit A — weaken the standing hypothesis

File `sections/Preliminaries/LocalRamificationPreliminaries.tex`, replacing lines 45–55.

Currently:

For the remainder of this section, assume that A is a discrete valuation ring of equal characteristic $p > 0$, complete with respect to its maximal-ideal-adic topology, with uniformizer π and **perfect residue field** κ . Thus $\text{char}(A) = \text{char}(\kappa) = p$. The Cohen structure theorem provides a coefficient field $k \subset A$ mapping isomorphically onto κ . Consequently, $A \cong k[[\pi]] \cong k[[t]]$, and its fraction field is $K \cong k((\pi))$. Let L/K be a finite Galois extension with Galois group G . The integral closure B of A in L is itself a complete discrete valuation ring. Since the residue field κ is perfect, the extension of residue fields is separable, and there exists a uniformizer $\Pi \in B$ such that $B = A[\Pi]$.

Proposed:

For the remainder of this section, assume that A is a discrete valuation ring, complete with respect to its maximal-ideal-adic topology (cf. [2, Tag 00IE] and [2, Tag 0324]), with uniformizer π , residue field κ and fraction field K . Let L/K be a finite Galois extension with Galois group G , and let B be the integral closure of A in L ; it is again a complete discrete valuation ring. We assume throughout that

$$\text{the residue extension } \kappa_B/\kappa \text{ is separable.} \tag{1}$$

This is the standing hypothesis of [1, Chapter IV], and it is what makes the constructions below available: under equation (??) there is an element $x \in B$ with $B = A[x]$ [1, Chapter III, §6, Proposition 12] (one may take x to be a uniformizer Π when L/K is totally ramified). Condition equation (??) holds automatically whenever κ is perfect, and in particular when κ is finite or algebraically closed; but it is strictly weaker, and the extra generality is what allows us to speak of ramification at the generic point of an exceptional divisor, whose residue field is a function field and hence imperfect in characteristic p .

In the equal-characteristic situation with κ perfect — the setting of Section 2.4 — one has more: if $\text{char}(A) = \text{char}(\kappa) = p > 0$, the Cohen structure theorem ([2, Tag 032A]; see also [2, Tag 0C0S]) provides a coefficient field $k \subset A$ mapping isomorphically onto κ , so that $A \cong k[[\pi]] \cong k[[t]]$ and $K \cong k((\pi))$.

Knock-on changes in the same section. Two, both mechanical.

- The definition of i_K^L is currently phrased through a uniformizer Π with $B = A[\Pi]$. Under the weaker hypothesis the monogenic generator need not be a uniformizer, so it becomes $i_K^L(\sigma) = v_B(\sigma(x) - x)$ for the element x with $B = A[x]$. Nothing downstream depends on x being a uniformizer: what is used is that $G_i = \{\sigma : i_K^L(\sigma) \geq i + 1\}$.
- The parenthetical “(complete local fields with a discrete valuation and perfect residue field)” becomes “... and separable residue extension”.

Safe downstream? Yes. The section on the cyclic theories re-declares its own hypothesis (“let K be a complete discrete valuation field with perfect residue field κ of characteristic $p > 0$ ”), so the Artin–Schreier–Witt and Kummer computations keep the explicit $k((t))$ description they need.

Edit B — the general definition

File `sections/Preliminaries/GTorsorRamification.tex`, inserted after [Definition 38](#) ends (line 24) — that is, *before* “Now back to our original assumptions”.

The point of the placement: the section already sets up X , U , (Z, ξ) in the right generality and then abandons it. This edit finishes the general setup before descending to the curve, rather than adding a new subsection.

Proposed:

[Definition 38](#) is qualitative. To measure ramification we specialize to torsors and pass to the completion. Keep X , U and (Z, ξ) as above, let G be a finite abelian group, and let $\mathcal{P} \rightarrow U$ be a G -torsor in $U_{\text{ét}}$. Write $\widehat{\mathcal{O}}_{X, \xi}$ for the completion of $\mathcal{O}_{X, \xi}$ and $\widehat{K}_\xi = \text{Frac}(\widehat{\mathcal{O}}_{X, \xi})$ for the associated complete discretely valued field. Restricting $\mathcal{P}$ to the punctured formal neighbourhood $\text{Spec}(\widehat{K}_\xi)$ isolates the ramification at (Z, ξ) : this restriction is finite étale over $\widehat{K}_\xi$, hence by [\[2, Tag 02GL\]](#) decomposes into a finite disjoint union of spectra of finite separable field extensions,

$$\mathcal{P}|_{\text{Spec}(\widehat{K}_\xi)} \cong \bigsqcup_{i=1}^N \text{Spec}(F_i),$$

with the following properties:

1. The extensions $F_i/\widehat{K}_\xi$ are mutually isomorphic and Galois, and each F_i is complete for the unique extension of the discrete valuation on $\widehat{K}_\xi$ [\[1, Chapter II, §2, Proposition 3\]](#).
2. If F denotes any one of the F_i , then $H = \text{Gal}(F/\widehat{K}_\xi)$ identifies with the stabilizer of the corresponding connected component of $\mathcal{P}|_{\text{Spec}(\widehat{K}_\xi)}$, which is a subgroup of G .
3. The number of components is $N = [G : H]$.

Definition 1. In the situation above, assume that the residue extension of $F/\widehat{K}_\xi$ is separable, so that the upper-numbering filtration of [Section 2.1](#) is defined for $F/\widehat{K}_\xi$; we then call (Z, ξ) *residually separable for $\mathcal{P}$* . We say that the extension $F/\widehat{K}_\xi$ *has ramification bounded by r* if the ramification group H^r is trivial. Here H^r denotes the upper-numbering ramification group of $H = \text{Gal}(F/\widehat{K}_\xi)$ associated with the extension $F/\widehat{K}_\xi$; it is not a filtration inherited from the ambient group G . This condition is independent of the chosen connected component of $\mathcal{P}|_{\text{Spec}(\widehat{K}_\xi)}$, and we say $\mathcal{P}$ *has ramification bounded by r at (Z, ξ)* when it holds.

It is convenient to restate this through characters. Since G is a finite abelian group over k , hence étale, the correspondence between G -torsors and the fundamental group ([Corollary 14](#))

applies; over the complete field $\widehat{K}_\xi$ the fundamental group is the absolute Galois group $G_{\widehat{K}_\xi} := \text{Gal}(\widehat{K}_\xi^{\text{sep}}/\widehat{K}_\xi)$, so that $\mathcal{P}|_{\text{Spec}(\widehat{K}_\xi)}$ corresponds to a unique continuous homomorphism

$$\rho_\xi : G_{\widehat{K}_\xi} \longrightarrow G,$$

with $H = \text{im}(\rho_\xi)$ and $F = (\widehat{K}_\xi^{\text{sep}})^{\ker \rho_\xi}$. Because the upper numbering is compatible with quotients, H^r is the image of $G_{\widehat{K}_\xi}^r$, and therefore

$$\mathcal{P} \text{ has ramification bounded by } r \text{ at } (Z, \xi) \iff \rho_\xi(G_{\widehat{K}_\xi}^r) = \{1\}. \quad (2)$$

Lemma 2. *Let (Z, ξ) be residually separable for $\mathcal{P}$. Then $\mathcal{P}$ has ramification bounded by 0 at (Z, ξ) if and only if it is unramified there, and bounded by 1 if and only if it is tamely ramified there, both in the sense of Definition 38.*

Proof. Ramification indices and residue extensions are unchanged under the faithfully flat extension $\mathcal{O}_{X, \xi} \rightarrow \widehat{\mathcal{O}}_{X, \xi}$, which induces an isomorphism on residue fields; so both conditions of Definition 38 may be tested after completion. Now $H^0 = H_0$ is the inertia group, which gives the first claim. For the second, $H_1 = 1$ implies $H^1 = 1$ directly. Conversely, since $H \subseteq G$ is abelian, the jumps of the upper-numbering filtration are integers by the Hasse–Arf theorem [1, Chapter V, §7, Theorem 1]; the jump at which H_1 dies therefore occurs at an integral upper index, so $H^1 = 1$ forces $H_1 = 1$. $\square$

Why the character paragraph is the load-bearing half. (2) is the form in which every proof in the subsection actually uses the notion. With it stated at a general ξ , the existing proofs of the contracted-product lemma, the étale-descent lemma, the fibre-product lemma and the induced-torsor lemma go through *without a single word changed* — they are already written in this language.

Edit C — collapse the curve paragraph

Same file, replacing lines 26–57 (the closed-point decomposition and the old *ramification bounded by r* definition), and deleting lines 155–173.

Currently:

Now back to our original assumptions, C is a projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z, ξ) is now a closed point $P \dots$ [followed by the decomposition $\mathcal{P}|_{\mathrm{Spec}(\hat{K}_P)} \cong \bigsqcup \mathrm{Spec}(F_i)$, its three properties, and the definition of *ramification bounded by r at P*].

Proposed:

Now back to our original assumptions: C is a projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z, ξ) is now a closed point P , the local ring $\mathcal{O}_{C,P}$ is a discrete valuation ring, and $\hat{K}_P = \mathrm{Frac}(\hat{\mathcal{O}}_{C,P})$ is the complete local field at P . The residual separability hypothesis of Definition 1 is automatic here, since $\kappa(P)$ is finite over the perfect field k and hence perfect. We write $\mathcal{P}$ has *ramification bounded by r at P* for the condition of Definition 1 applied to $(Z, \xi) = (\{P\}, P)$.

The two modulus definitions and the proposition comparing the closed-point and geometric formulations then follow *unchanged*.

Two deletions.

- The sentence “Furthermore, the notions of tame ramification and unramifiedness derived here coincide with the codimension-1 criteria in Definition 38” is now Lemma 2, with a proof.
- The paragraph beginning “There is another equivalent formulation through the language of characters” moves upstream into Edit B and is deleted here.

Edit D — pin the lemma hypotheses

Same file, the statements of the four stability lemmas.

The contracted-product lemma currently opens with a free-standing setup. It should point at the definition, exactly as the fibre-product lemma already points at it:

Proposed:

Let G be a finite abelian group, and let X , U and (Z, ξ) be as in [Definition 1](#), with (Z, ξ) [residually separable for both \$\mathcal{P}_1\$ and \$\mathcal{P}_2\$](#) . Assume $\mathcal{P}_1$ has ramification bounded by r_1 at (Z, ξ) and $\mathcal{P}_2$ has ramification bounded by r_2 at (Z, ξ) . Then their contracted product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ has ramification bounded by $\max(r_1, r_2)$ at (Z, ξ) .

The same clause is added to the étale-descent lemma, at both ξ_X and ξ_Y — with the remark that there residual separability at ξ_X *follows* from residual separability at ξ_Y , since f is étale at ξ_X , so the hypothesis need only be imposed downstairs. The fibre-product and induced-torsor lemmas already say “let $X, U, (Z, \xi)$ be as in” the contracted-product lemma and so inherit it automatically.

Consequences, and three things to decide

Numbering. Adding [Lemma 2](#) as a numbered statement shifts everything after it by one: the contracted-product lemma moves from Lemma 43 to Lemma 44, and so on to the end of the thesis. Every reference is `\Cref`’d, so this recompiles correctly, but “Lemma 43” stops meaning what it means today. The alternative is to set it as a `rem*` — your `remarkstyle` is `numbered=no` — which preserves all numbering at the price of not being referenceable.

Where residual separability must now be checked. The hypothesis is genuinely needed at η_n , and it is genuinely verified there: the Artin–Schreier argument shows each factor of $B \otimes_A \widehat{K}_\eta$ has “ramification index 1 and separable residue extension”. What the edit changes is that this verification now discharges a stated hypothesis instead of being an unmotivated step. Worth a one-line forward reference from the definition to that computation.

One reference to fill in. The claim that ramification indices and residue extensions are unchanged by completion is argued in one sentence in the proof of [Lemma 2](#). It is elementary ($B \otimes_A \widehat{A} \cong \prod_i \widehat{B_{\mathfrak{m}_i}}$, preserving e and f), but if you would prefer a citation there I would rather look up the Stacks tag than guess at one.

Alternative worth a moment’s thought. Everything above keeps “bounded by r ” anchored to the upper-numbering filtration of the finite quotient H . One could instead take equation (??) — the condition $\rho_\xi(G_{\widehat{K}_\xi}^r) = 1$ on the absolute Galois group — as the *definition*, and derive the H^r description as a consequence. That reads more cleanly, matches Guignard’s formulation, and makes the stability lemmas nearly tautological; the cost is that it obscures the elementary content and makes the comparison with [Definition 38](#) less direct. I did not take it, but it is a real choice and it is easy to switch.

Bibliography

- [1] J.-P. Serre, *Corps Locaux (Local Fields)*, Hermann / Springer GTM 67.
- [2] The Stacks Project Authors, *The Stacks Project*, <https://stacks.math.columbia.edu>.
- [3] Q. Guignard, *Geometric local class field theory*.